Final Project Report

Project Name:

Ecosystem Discovery Application (EcoDiscovery)

ENSE 281- Software Engineering

University of Regina

Team Name and Team Members:

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Introduction:

The Ecosystem Discovery Application is a web-based educational game designed for children between second to third grade. The primary goal of the application is to introduce young learners to freshwater ecosystems found in Saskatchewan while promoting environmental awareness through interactive gameplay. The system allows users to explore aquatic animals, interpret simple hints, and engage with conservation-focused mechanics such as cleaning trash from the lake.

This application consists of four main pages: a home page, a game page, a video page, and an about us page. Among these, the game page serves as the core component of the application, where users interact with the ecosystem and complete learning tasks. The project was developed using HTML, CSS, and JavaScript, structured with a Model-View-Controller (MVC) architecture, and supported by Supabase for managing dynamic data.

Project Goals and Objectives:

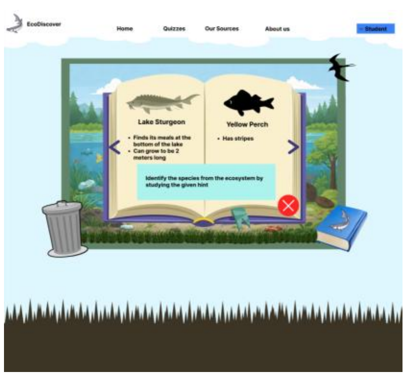
The main objective of this project was to design and implement an engaging, interactive learning tool tailored to young children. Specifically, the team aimed to create a system that combines education and gameplay in a way that is both intuitive and enjoyable for the target age group. The application was intended to teach users about freshwater animals while also introducing basic environmental responsibility concepts.

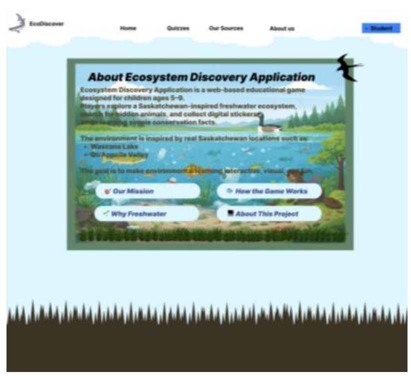
Another key goal was to implement a structured software design using the MVC architecture. This was intended to promote separation of concerns, improve maintainability, and allow different team members to work on independent components of the system. Additionally, the team aimed to ensure that the user interface was accessible, visually appealing, and appropriate for children with limited reading ability.

Overall, the project successfully achieved its intended goals. The final application provides a clear and engaging user experience, integrates educational content effectively, and demonstrates a well-organized technical structure.

Design Process and Initial Prototyping:

The design process began with a low-fidelity prototype, using rough sketches on paper and board during several group meetings, and then the creation of a high-fidelity prototype using Figma. This prototype was used to explore layout options, visual styles, and interaction flows before implementation. The initial design featured a detailed sticker book interface alongside a visually rich ecosystem environment, with multiple UI elements providing descriptive information about each animal.





However, during early feedback sessions, several issues were identified. The interface was found to contain too much text for the intended age group, making it potentially overwhelming for young users. Additionally, some text elements were difficult to read due to insufficient contrast with background colours. There were also concerns about the amount of time children might spend interacting with the application, as well as suggestions to ensure that the project remained within its intended scope.

Incorporation of Feedback:

The feedback received during the prototyping phase had a significant impact on the final design of the application. One of the most important changes involved reducing the amount of text displayed on the screen. Long descriptions were replaced with short, simple hints and concise facts that are easier for children to understand. This shift ensured that the application relies more on visual interaction rather than extensive reading.

To address visibility concerns, the design was updated to improve contrast between text and background elements. Text was placed within clearly defined UI components, such as cards, rather than directly on complex backgrounds. Font sizes were also increased to improve readability.

In response to the concerns about screen time, a global timer system was introduced. This timer limits the duration of gameplay and encourages users to engage with the application in short, focused sessions. Finally, the team carefully reviewed all features to ensure they remained within the project scope, removing unnecessary complexity and focusing on core functionality.

System Architecture and Implementation:

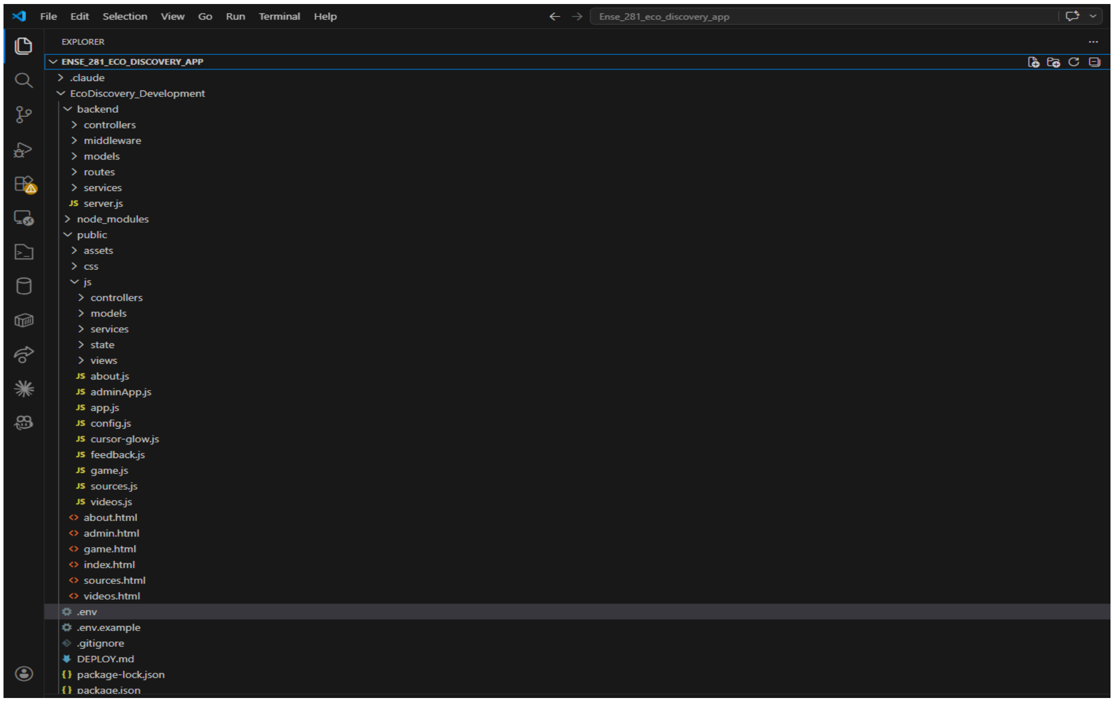
The application was built using a Model-View-Controller (MVC) architecture with a JavaScript frontend and a Node.js/Express backend. These two parts communicate through a REST API, which means the frontend does not directly access the database. This helps keep the system more secure and easier to manage.

On the frontend, the code is organized into models, views, and controllers. The model handles data and game state, the view is responsible for displaying the interface, and the controller manages user interactions and game logic. A central state file keeps track of the game progress, while an API service handles requests to the backend.

The backend acts as a bridge between the frontend and the database. It defines API routes, processes requests, and retrieves data using model functions. Middleware such as CORS and security tools are used to handle requests safely. The backend also serves the frontend files so the entire application can run from one server.

The database is managed using Supabase, which provides a PostgreSQL database and authentication features. All data, such as animals and hints, is accessed through the backend, and sensitive information is stored securely using environment variables.

Overall, this structure keeps the system organized, secure and easy to maintain.



Core Gameplay and User Interaction:

The central feature of the application is the interactive gameplay experience on the game page. The gameplay is structured around a sequence of tasks in which the user must identify freshwater animals based on simple hints. When the user selects a mystery sticker, a hint is revealed, prompting the user to locate the correct fish within the lake environment.

If the user selects the correct animal, the corresponding sticker is unlocked, and a short educational fact is displayed. If the user selects an incorrect animal, the system provides immediate feedback, encourages the user to try again, and provides an additional hint. The user is given up to three tries to guess the intended animal, before it is locked for that round. This interaction loop promotes active learning by requiring the user to interpret hints and make decisions based on observation.

The gameplay continues until all stickers have been collected and/or locked or the timer runs out. This structure creates a sense of progression and achievement while maintaining a clear educational focus.

Conservation Feature - Lake Cleanup Mechanic:

To reinforce the environmental theme of the application, a conservation feature was introduced in the form of a lake cleanup mechanic. In this feature, trash items appear within the lake environment, and users can drag them into a garbage bin.

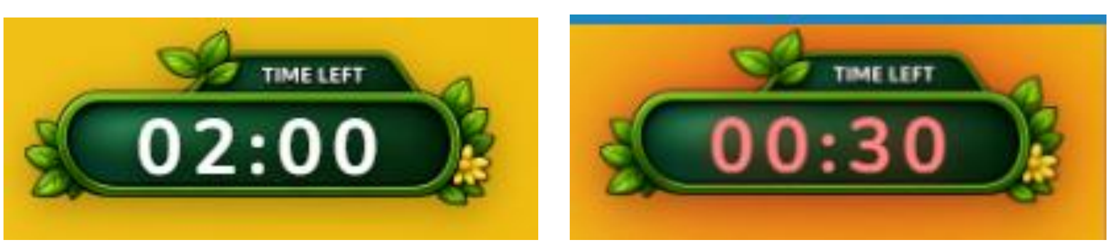
Each correctly disposed trash item rewards the user with additional time, specifically 20 seconds per item, with a maximum of three items per game. This creates a direct link between environmental responsibility and gameplay benefits. The inclusion of this feature serves both an educational and motivational purpose. It teaches users the importance of keeping natural environments clean while also providing a meaningful incentive to engage with the feature.

Timer System:

The timer system was redesigned during development to better support the overall gameplay experience. Instead of assigning a separate timer for each sticker, the final design uses a single global timer of five minutes for the entire game session.

This approach creates a more cohesive experience by encouraging users to manage their time strategically across all tasks. It also increases the level of challenge, as users must balance exploration, decision-making, and optional activities such as cleaning trash.

The addition of bonus time through the cleanup mechanic further enhances this system by providing players with opportunities to extend their gameplay through positive actions.

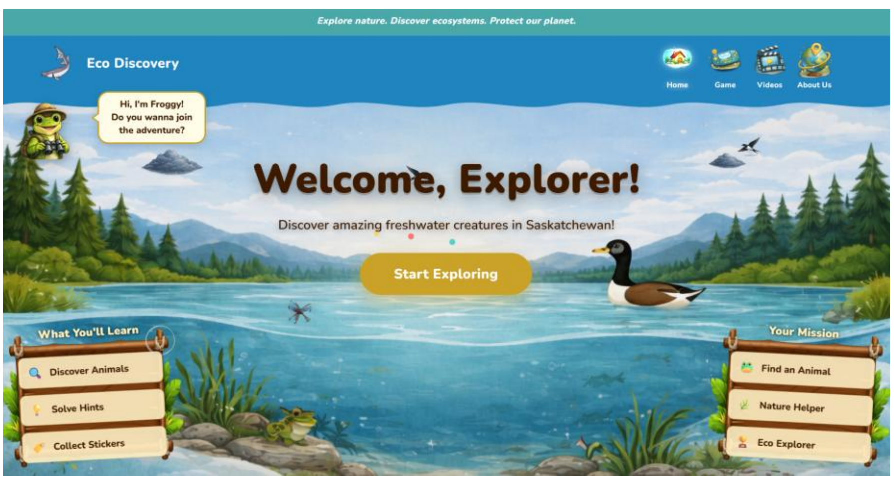


Final Outcome

The look of the final product is significantly different from that of our original prototypes. This is due to design changes that were made based on peer, user, and professor feedback; ease of implementation; and more thought-out and thorough ideas regarding site utility and gameplay. A key change that was made was ensuring that site text could be understood and read by the second to third grade children the site is aimed towards. To accomplish this, we utilized the Hemingway Editor's readability checker that our professor recommended. It helped us to ensure a general maximum reading level of grade three. Some exceptions concerning more difficult-to-read words were made to preserve what we thought was important, such as "Saskatchewan", "freshwater", and the names of animals. We were also told to ensure our designs were fun and child friendly, to potentially include less text, and to simplify certain images. This advice was kept in mind for our redesigns. The whole site is themed around freshwater rivers, lakes, and animals. The navigation bar is flowing water, and each page features an ecosystem image and moving creatures. The chosen colours are also bright, fun, and saturated, providing visual interest.

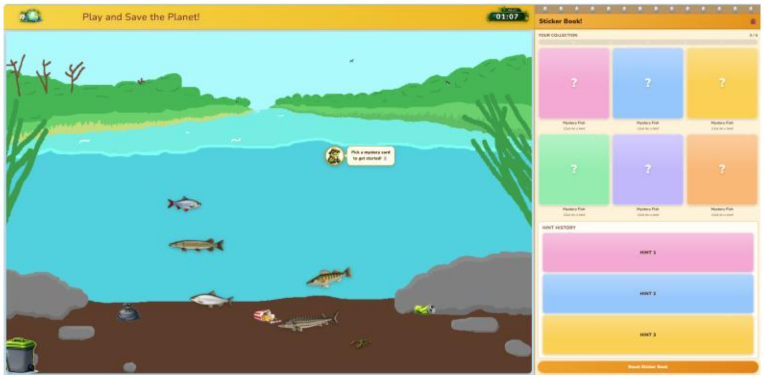
The final site includes a home page, game page, videos page, and about us page.

Home Page



The website's home page introduces the user to the concept and usefulness of our product. The welcome text invites the user to explore the featured game and information cards detail what the user can do with and expect to find on our product. During a design demonstration, we got feedback on our previous navigation bar stating that it was not fun and engaging enough for young children. To rectify this, we added images to represent the different navigation options. The original high-fidelity prototype also represented the navigation bar as a cloud. In the final design, this was changed to a wave to better represent freshwater ecosystems, as well as due to its easier implementation. Moving clouds and birds were also added to enhance visual interest and the “Start Exploring” button grows and shrinks guiding the user towards entering the game.

Game Page



Substantial changes were made to the design of the game page. We kept the original idea of a section view of an ecosystem with swimming fish and garbage collection. However, we made the sticker book a bit more abstract. Instead of an image of an actual book with turntable or symbolic pages, we made it a grid of unlockable boxes. Each of these boxes represent an unlockable sticker. This was done for two reasons. The first one being that we wanted the user to be able to see the hints while they are searching for the correct animal. This was not possible with the original design. We also wanted the user to be able to select the animal they want to find and for there to be visual feedback when an incorrect selection is made. This was, again, more difficult to do with a toggleable book with tunable pages. We also increased the number of hints from one to three. This allows the user to learn more about the animals. A fun progress tracker was also added to add a sense of achievement.

Video Page

To provide a greater opportunity for learning we also included a videos page. This replaced our previous quiz page. We wanted to provide a space for children to comfortably learn more about freshwater ecosystems, its animals, and nature in general. These videos are a great addition since they were already made for children and are potentially more engaging than simply reading text. They also allow children to learn information without needing to be able to read it.

About Us Page

The about us page is made for the parents, guardians, and teachers of the children, as well as for anyone who may want more information about our site and why it exists. It provides a mission statement, our values, and the names of our team members.

Challenges and Problem Solving:

Throughout the development process, the team encountered several challenges. One of the main challenges was the initial overcomplication of the design, particularly in terms of excessive text and visual clutter. This was addressed by simplifying the interface and focusing on essential elements.

Another challenge involved the use of phaser as a game engine. While Phaser provided powerful features, it introduced additional complexity and created dependencies between team members. As a result, the team decided to remove Phaser and implement the interaction using simpler JavaScript techniques, which improved stability and reduced development overhead.

Team coordination also presented challenges, particularly when integrating different components. The use of the MVC architecture helped mitigate this issue by clearly defining responsibilities and allowing team members to work independently on different parts of the system.

Reflection

This project was both challenging and rewarding for the team. At the beginning, the team had ambitious ideas and attempted to include many features, which led to an overly complex design. Over time, we learned that simplicity is often more effective, especially when designing for young users.

One of the most valuable lessons learned was the importance of listening to feedback. Initially, it was difficult to accept that parts of our design needed to be simplified or removed.

However, applying feedback significantly improved the usability and overall quality of the application.

We also experienced challenges with tools and technical decisions, particularly with the use of Phaser. Letting go of that approach and switching to a simpler implementation was not easy, but it ultimately made the project more manageable and aligned better with the project scope.

As a team, we learned the importance of communication and collaboration. There were moments of confusion and frustration, especially when integrating different parts of the system, but working through these issues helped us improve our teamwork and problem-solving skills.

Individually, this project helped us better understand our strengths and areas for improvement. Some team members developed stronger technical skills, while others improved in design thinking and organization. Overall, the experience has prepared us to approach future projects with a more balanced and user-centered mindset.

Conclusion:

The Ecosystem Discovery Application successfully demonstrates how an educational tool can combine interactive gameplay with meaningful learning outcomes. By focusing on simplicity, usability, and environmental awareness, the team was able to create an application that is both engaging and educational for young users.

The project highlights the importance of iterative design, responsiveness to feedback, and structured development practices. These lessons will be valuable in future projects and provide a strong foundation for continued growth in software development and user experience design.